

HARBIY XAVFSIZLIK VA MUDOFAA UNIVERSITETINING AXBOROT-KOMMUNIKATSIYA TEXNOLOGIYALARI VA HARBIY ALOQA INSTITUTI

2-MAVZU: PyQt5 paketi va QtDesigner dasturi yordamida GUI dasturlarini yaratish

O‘qituvchi: “Axborot texnologiyalari va sun’iy intellekt” kafedrası “Sun’iy intellekt” sikli katta o‘qituvchisi **katta leytenant Qodomboyev R.O‘.**

Toshkent 2026

2-mavzu. 3-mashgʻulot (Guruhli):
PyQt5 kutubxonasi. QLineEdit vidjeti.

O'quv savollari:

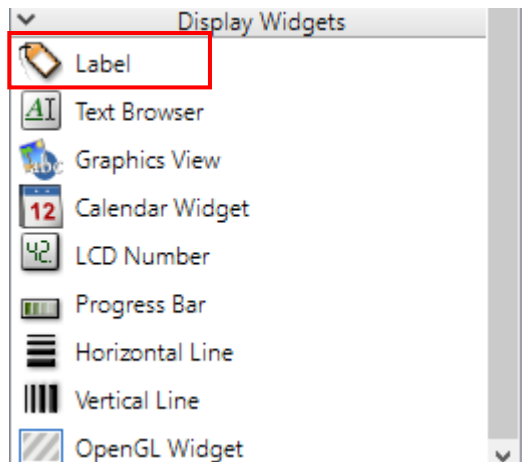
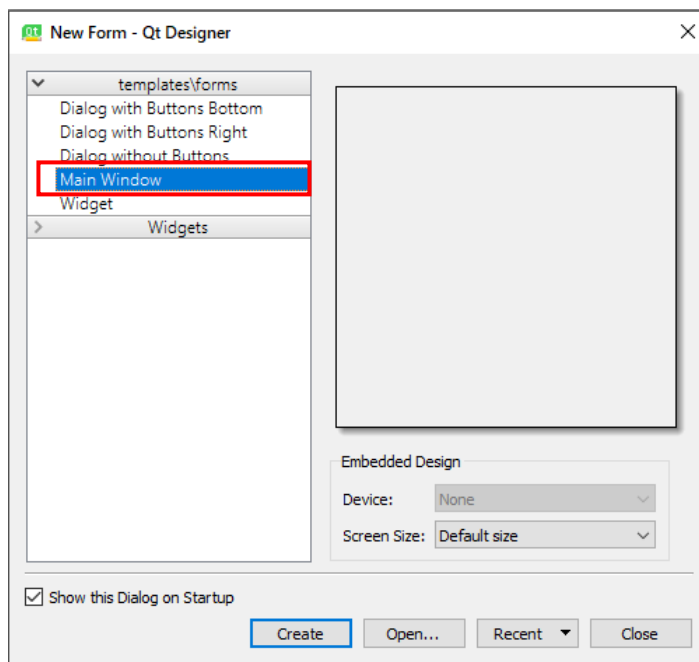
1. QLineEdit vidjeti.
2. setStyleSheet() metodi.

QLineEdit vidjeti:

1. Qt Designerda .ui fayl yaratish

Qadamlar:

- Qt Designerni oching.
- “Main Window” yoki “Dialog without Buttons” tanlang.
- Chap tomondan **Widget Box** → **Display Widgets** → **Label** ni oyna ichiga olib chiqing.
- Labelning `objectName` xususiyatini (Properties oynasida) **label_hello** deb o‘zgartiring.
- Faylni **main.ui** nomi bilan saqlang.



Property	Value
QObject	
objectName	label_hello
QWidget	
enabled	☒
geometry	[(280, 220), 47 x 13]
sizePolicy	[Preferred, Preferred, 0, 0]
minimumSize	0 x 0
maximumSize	16777215 x 16777215
sizeIncrement	0 x 0
baseSize	0 x 0
palette	Inherited
font	A [MS Shell Dlg 2, 8]

2. .ui faylni Python kodda yuklash

Biz .ui faylni ikkita yo‘l bilan ishlatishimiz mumkin:

1-usul: uic.loadUiType() orqali to‘g‘ridan-to‘g‘ri yuklash

Bu usul .ui faylni to‘g‘ridan-to‘g‘ri ochadi, **hech qanday konvertatsiya kerak emas.**

```
from PyQt5 import uic
from PyQt5.QtWidgets import QApplication, QMainWindow
import sys

class MyApp(QMainWindow):
    def __init__(self):
        super().__init__()
        uic.loadUi("main.ui", self)  # UI faylni yuklash
        # QLabel vidjetini ID orqali chaqirish
        self.label_hello.setText("Salom, PyQt5 va Qt Designer!")

app = QApplication(sys.argv)
win = MyApp()
win.show()
sys.exit(app.exec_())
```

2-usul: .ui faylni .py faylga aylantirish

Agar .ui faylni kod shaklida ishlatmoqchi bo'lsangiz, pyuic5 yordamida uni .py ga o'tkazasiz:

```
pyuic5 -x main.ui -o main_ui.py
```

Keyin main_ui.py faylni dasturda import qilib ishlatish mumkin:

```
from PyQt5.QtWidgets import QApplication, QMainWindow
from main_ui import Ui_MainWindow # .ui dan o'tgan kod
import sys
```

```
class MyApp(QMainWindow):
    def __init__(self):
        super().__init__()
        self.ui = Ui_MainWindow()
        self.ui.setupUi(self)
        self.ui.label_hello.setText("Qt Designer bilan ishladik!")

app = QApplication(sys.argv)
win = MyApp()
win.show()
sys.exit(app.exec_())
```

QLabel nima?

QLabel — bu **PyQt5 kutubxonasidagi** eng oddiy, lekin eng ko‘p ishlatiladigan **vidjetlardan (elementlardan)** biridir. U **matn (text)** yoki **rasm (image)** ko‘rsatish uchun ishlatiladi.

Ya’ni, QLabel — bu **ko‘rsatish (display)** uchun mo‘ljallangan vidjetdir, foydalanuvchidan ma’lumot olmaydi, faqat ma’lumotni chiqaradi.

QLabel vazifalari

QLabel quyidagi asosiy vazifalarni bajaradi:

1. Matnni ko‘rsatish

Oddiy yoki formatlangan (HTML) matnni chiqara oladi.

```
label.setText("Salom, PyQt5!")
```

2. Shrift va o‘lchamni o‘zgartirish

Matn shriftini, o‘lchamini va rangini o‘zgartirish mumkin:

```
from PyQt5.QtGui import QFont  
label.setFont(QFont("Arial", 14))
```

3. Rasm (Pixmap) koʻrsatish

QLabel yordamida rasmni ekranga chiqarish oson:

```
from PyQt5.QtGui import QPixmap  
label.setPixmap(QPixmap("rasm.png"))
```

4. HTML yoki link koʻrsatish

QLabel HTML formatni tan oladi:

```
label.setText('<a href="https://python.org">Python saytiga oʻtish</a>')  
label.setOpenExternalLinks(True)
```

5. Matn joylashuvini boshqarish

Matnni chapga, oʻrtaga yoki oʻngga tekislash mumkin:

```
from PyQt5.QtCore import Qt  
label.setAlignment(Qt.AlignCenter)
```

6. Chegara (ramka) va fon berish

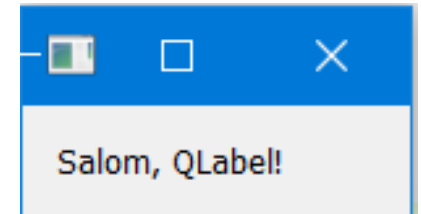
CSS uslubida style berish mumkin:

```
label.setStyleSheet("border: 1px solid gray; background-color: #eef;")
```


Misollar

1-misol. QLabel yaratish

```
from PyQt5.QtWidgets import QApplication, QLabel, QWidget, QVBoxLayout
import sys
app = QApplication(sys.argv)
win = QWidget()
layout = QVBoxLayout(win)
label = QLabel("Salom, QLabel!")
layout.addWidget(label)
win.show()
sys.exit(app.exec_())
```



2-misol:

```
from PyQt5.QtWidgets import QApplication, QLabel, QWidget, QVBoxLayout
from PyQt5.QtGui import QFont, QPixmap
from PyQt5.QtCore import Qt
import sys

app = QApplication(sys.argv)

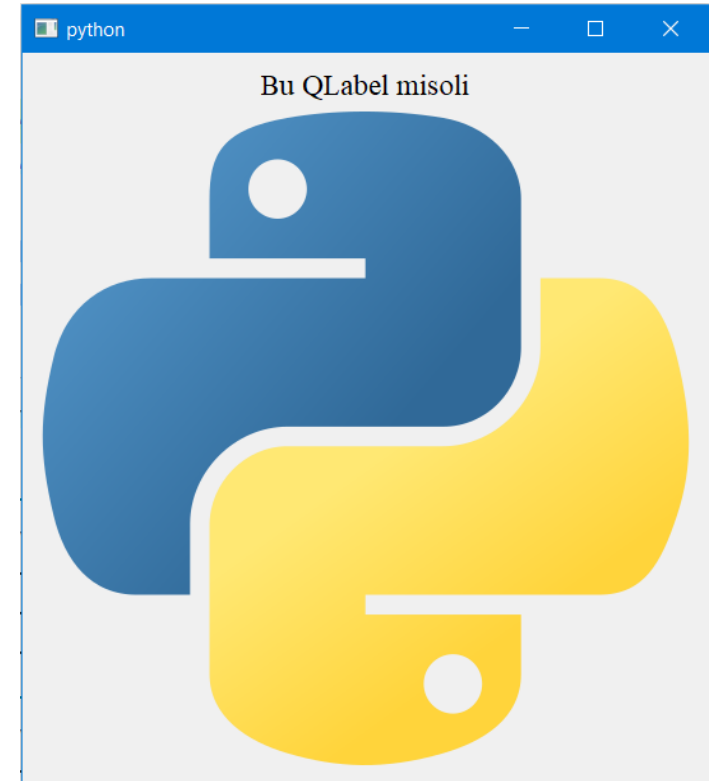
win = QWidget()
layout = QVBoxLayout(win)

# Matnli Label
label1 = QLabel("Bu QLabel misoli")
label1.setFont(QFont("Times New Roman", 14))
label1.setAlignment(Qt.AlignCenter)

# Rasmlı Label
label2 = QLabel()
label2.setPixmap(QPixmap("python_logo.png"))
label2.setScaledContents(True)

layout.addWidget(label1)
layout.addWidget(label2)

win.show()
sys.exit(app.exec_())
```



2. setStyleSheet() metodi.

1. Asosiy: matn (text) bilan ishlash

```
label.setText("Salom!")           # QLabel ga matn yozish yoki mavjud matnni yangilash
current = label.text()            # joriy matnni olish
```

- setText() — matn yozish yoki mavjud matnni yangilash.
- text() — joriy matnni qaytaradi.

2. Oddiy matn va HTML (Rich text)

```
from PyQt5.QtCore import Qt

label.setTextFormat(Qt.PlainText)  # qat'iy oddiy matn
label.setText("1 < 2 & 3")         # PlainText bo'lsa HTML teglari ishlamaydi
label.setTextFormat(Qt.RichText)   # HTML (rich text) ishlaydi
label.setText('<b>Qalin</b> va <span style="color:red">qizil</span>')
label.setOpenExternalLinks(True)  # <a href="..."> linklarini tashqi brauzerda ochish
```

- Qt avtomatik aniqlaydi, lekin kerak bo'lsa .setTextFormat() bilan aniq belgilash mumkin.

3. Font (shrift) — QFont yordamida

```
from PyQt5.QtGui import QFont
f = QFont("Arial", 14)      # oilasi va point-size (nuqta o'lchami)
f.setBold(True)            # qalin
f.setItalic(True)          # kursiv
f.setUnderline(True)       # ostiga chizish
f.setWeight(75)            # vazn: 0-99 (normal ~50, bold ~75)
label.setFont(f)
```

- QFont(family, pointSize) — family (masalan "Arial"), pointSize — nuqtada.
- setWeight(int) aniq nazorat uchun.
- setPixelSize(px) — agar piksel asosida aniq o'lchash kerak bo'lsa:
- f.setPixelSize(20)

4. Style sheet (CSS uslubida) bilan font va rang

```
label.setStyleSheet("font-family: 'Times New Roman'; font-size: 18pt; color: #2a7ae2;")
```

- font-size da pt yoki px ishlatish mumkin (18pt, 16px).
- Style sheet yordamida yana font-weight, font-style va boshqalarni ham berish mumkin:
- label { font-weight: bold; font-style: italic; }

5. Matn joylashuvi, margin, indent va o'rash

```
from PyQt5.QtCore import Qt  
label.setAlignment(Qt.AlignCenter)    # AlignLeft, AlignRight, AlignTop, AlignVCenter ...  
label.setMargin(8)                   # ichki chekka (pixel)  
label.setIndent(10)                  # matnni chapdan surish  
label.setWordWrap(True)              # uzun matnni yangi qatorga o'rash
```

- setWordWrap(True) — matn label kengligiga sig'masa avtomatik yangi qatorga o'tadi.

6. Matn tanlash va interaktivlik (link, select)

```
from PyQt5.QtCore import Qt
label.setTextInteractionFlags(Qt.TextSelectableByMouse)    # sichqoncha bilan tanlash mumkin
label.setTextInteractionFlags(Qt.TextBrowserInteraction)  # link va tanlash kombinatsiyasi
# Link signali:
label.setText('<a href="hello">Bos!</a>')
label.linkActivated.connect(lambda href: print("Link bosildi:", href))
```

- linkActivated(str) — ichki link bosilganda chaqiriladi.
- setOpenExternalLinks(True) — tashqi linklarni brauzerda ochadi.

7. Matnni labelga sig'dirish: QFontMetrics va elide

Agar matn labelga sig'masa, uni ... bilan qisqartirish:

```
from PyQt5.QtGui import QFontMetrics
from PyQt5.QtCore import Qt
fm = QFontMetrics(label.font())
elided = fm.elidedText("Bu juda uzun matn...", Qt.ElideRight, label.width())
label.setText(elided)
```

•Qt.ElideRight — oxiridan ... qo'yadi. ElideLeft, ElideMiddle ham bor.

8. Avtomatik shrift moslash (fit-to-width) — kod misoli

Bu funktsiyani oynani ko'rsatgandan yoki o'lchami o'zgarganda chaqiring.

```
def fit_font_to_label(label, text, min_size=6, max_size=48):  
    from PyQt5.QtGui import QFont, QFontMetrics  
    from PyQt5.QtCore import Qt  
    for size in range(max_size, min_size-1, -1):  
        f = QFont(label.font().family(), size)  
        fm = QFontMetrics(f)  
        if fm.horizontalAdvance(text) <= label.width():  
            label.setFont(f)  
            label.setText(text)  
            return  
  
    # agar hech qaysi sigmasa, elide qilamiz  
    fm = QFontMetrics(label.font())  
    label.setText(fm.elidedText(text, Qt.ElideRight, label.width()))
```

- Bu label kengligiga mos holda eng katta shriftni topadi.

9. Rasm bilan ishlash va shrift bilan bog'liq (pixmap)

Agar QLabel rasm ko'rsatsa:

```
from PyQt5.QtGui import QPixmap
```

```
label.setPixmap(QPixmap("img.png"))
```

```
label.setScaledContents(True)    # Label o'lchamiga moslab o'zgartiradi
```

- `setScaledContents(True)` rasmni label o'lchamiga moslashtiradi — sifatga e'tibor bering.

